國立東華大學應用數學系

碩士論文

時空 skew-t 過程於門檻超標預測之延伸：AR(2) 時間先驗與多尺度樣條基底函數

Extending the Spatio-Temporal Skew-t Process for Threshold Exceedance Prediction: AR(2) Temporal Priors and Multi-Resolution Spline Basis Functions

研究生：謝易軒
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摘要

在環境風險治理中，決策關心的通常不是平均濃度，而是未設站位置超越高門檻的機率。本文以 Morris et al. (2017) 的時空 skew-t 過程 (STP) 爲基準，沿兩軸檢驗其可改進空間：潛在參數的標準化 AR(2) 先驗，與迴歸均值中的多尺度薄板樣條 (MRTS) 基底。模型比較以樣本外 Brier 分數爲核心指標，於模擬與 2005 年 7 月美國 AQS 臭氧資料（CMAQ 共變數）上進行。時間軸的結果依預測任務而分：即使係數被完整回收，空間內插仍對自迴歸階數不敏感；只有當任務需要外推至觀測紀錄之外，較豐富的時間 pooling 才帶來從未顯著的增益。MRTS 均值項在非定態確實位於均值曲面、且基底足以解析它時可改善預測；於臭氧資料上其增益真實，惟幅度不大。因此本文的貢獻在於將 STP 基準擴展爲可由樣本外門檻預測表現進行資料驅動選擇的候選模型類。

關鍵字：時空 skew-t 過程、門檻超標預測、AR(2) 先驗、多尺度薄板樣條、Brier 分數、臭氧

Abstract

In environmental risk regulation, the inferential target is often not the field mean but the probability of exceeding a high threshold at unmonitored locations. We take the spatio-temporal skew-t process (STP) of Morris et al. (2017) as a baseline and examine where predictive improvement remains possible along two axes: a standardized AR(2) prior on the latent parameters and multi-resolution thin-plate spline (MRTS) basis functions in the regression mean. Models are compared by held-out Brier scores in simulation and on July 2005 U.S. AQS ozone data with CMAQ covariates. The temporal axis divides along the prediction task: spatial interpolation is insensitive to the autoregressive order even where the coefficients are recovered closely, and richer temporal pooling helps only when the task requires extrapolating beyond the observed record, by margins never significant. The MRTS mean improves prediction where the non-stationarity lies in a genuine mean surface and the basis is fine enough to resolve it; on the ozone data its gain is real, though modest in magnitude. The contribution is therefore to enlarge the STP baseline into a candidate model class selected by held-out threshold predictive performance.

Keywords: spatio-temporal skew-t process; threshold exceedance prediction; AR(2) prior; multi-resolution thin-plate splines; Brier score; ozone

Contents

1 Introduction

Demand for accurate pollution-concentration estimates over broad spatial domains continues to grow, driven by epidemiological evidence tying air quality to morbidity and mortality and by tightening regulation, yet the cost and logistics of a monitoring network cap the coverage that direct measurement can provide. This gap motivates statistical methods that extend a network's spatial reach with quantified confidence (U.S. Environmental Protection Agency, 2024). Ground-level ozone is the canonical regulatory case. The National Ambient Air Quality Standard (NAAQS) specifies the maximum concentration of ozone (O_3) allowed in outdoor air. A site complies only if the three-year average of its annual fourth-highest daily maximum eight-hour concentration stays at or below the current NAAQS of 0.070 parts per million (70 ppb), last revised in 2015. The Clean Air Act requires the EPA to review the standard at least every five years; Table 1 lists its successive revisions. The policy-relevant target is thus not the mean field but its upper tail, namely the probability that an unmonitored location breaches a high threshold; our goal is to predict that probability at unmonitored sites and to map it across the domain.

Table 1: Ozone National Ambient Air Quality Standard

Form of the Standard (parts per million, ppm)	**1997**	**2008**	**2015**
Annual 4^{th} highest daily max 8-hour average, averaged over three years	0.080	0.075	0.070

A spatial model for ozone threshold exceedances warrants special consideration, since standard spatial methods are likely to perform poorly. Two features of the problem are decisive. First, likelihood-based spatial modeling typically assumes a Gaussian process, which is appropriate when interest lies in mean behavior. The Gaussian distribution is light-tailed and symmetric. Daily ozone maxima are neither. Their residuals are heavy-tailed and are well described by a skew-t distribution (Morris, Reich, Emeric Thibaud, and Cooley, 2017). Second, our interest is in the dependence structure at high levels, which a covariance designed for mean behavior does not capture. A Gaussian process is asymptotically independent outside the degenerate case of perfect dependence, so the probability that two sites are jointly extreme decays to zero as the level increases. Nearby ozone monitors, however, tend to record high values on the same days, and this agreement weakens with distance (Morris, Reich, Emeric Thibaud, and Cooley, 2017). These considerations call for a heavy-tailed, asymmetric margin that retains asymptotic dependence. The skew-t distribution provides both. It is a subasymptotic model, capturing the joint tail decay at high but finite levels rather than the limiting dependence of extreme events (Raphaël Huser and Wadsworth, 2022). This is the setting the ozone data occupy.

Our approach also departs from threshold models based on extreme-value distributions, a well-developed part of the field in which extreme events are defined as exceedances over a high threshold. Anthony C. Davison and Smith (1990) model univariate exceedances with the generalized Pareto distribution. Ledford and J. A. Tawn (1996) address the multivariate case through a censored likelihood that distinguishes the ways in which a threshold may be exceeded. Wadsworth and J. A. Tawn (2012) and E. Thibaud, Mutzner, and A. C. Davison (2013) carried these ideas to spatial extremes, fitting models by a censored pairwise likelihood (Padoan, Ribatet, and Sisson, 2010). R. Huser and A. C. Davison (2014) extended them to space-time data. Such methods are grounded in extreme-value theory and presume a threshold high enough for extremal models to be valid and for extrapolation beyond the observed range. They are also computationally demanding and, in practice, restricted to small datasets. Neither feature suits the present application, since the 75 ppb threshold corresponds to roughly the 0.92 sample quantile of the daily ozone maxima (Morris, Reich, Emeric Thibaud, and Cooley, 2017), well short of the far tail that extremal models require, and the network, at $1{,}089$ monitoring stations, is far larger than such methods can handle. A complementary strategy retains the max-stable process while reducing its computational cost. Fully Bayesian inference for max-stable processes is cumbersome at scale. Morris, Reich, and Emeric Thibaud (2019) address this by truncating the spectral representation to a small set of empirical basis functions that carry the spatial dependence. Their construction renders inference feasible for large datasets and, as a by-product, dispenses with the assumption of stationarity.

In this thesis we build on the spatial skew-t process of Morris, Reich, Emeric Thibaud, and Cooley (2017). This process is not max-stable, yet it retains extremal dependence. Their model has four components. A skew-t margin, built by location-scale mixing of Gaussian processes, supplies heavy tails and asymmetry. A random spatial partition, redrawn each day, confines asymptotic dependence to nearby sites. A censored likelihood focuses inference on exceedances above a threshold. A transformed first-order autoregressive process carries dependence through time. We extend this baseline along two axes. On the temporal axis, the autoregressive prior links each day only to the one before, and we add a further lag to capture longer-range dependence in the record. On the spatial axis, the baseline assumes a stationary Matérn covariance, restrictive over a continental network. We therefore introduce the multi-resolution thin-plate spline basis of Tzeng and Huang (2018) as fixed-effect covariates in the mean, following the basis-function idea reviewed above, so as to relax stationarity at little additional cost. Each axis is a candidate extension rather than a replacement of the baseline, and together they are intended to improve prediction of threshold exceedances.

The remainder of the thesis is organized as follows. Section 2 reviews the spatial skew-t process of Morris, Reich, Emeric Thibaud, and Cooley (2017), develops the two extensions in turn (the standardized AR(2) prior on the latent dynamics and the MRTS basis functions in the mean), and assembles them into a single hierarchical model together with the MCMC algorithm used for inference. Section 3 evaluates each extension on synthetic data, describing the shared simulation design and the validation protocols, with the uncertainty attached to each score comparison, and reporting the AR(2) and MRTS experiments in turn. Section 4 applies the models to the ozone monitoring data, comparing the baseline against the two extensions in their ability to predict threshold exceedances. Section 5 concludes and outlines directions for further work. Technical derivations and design details are collected in the appendix.

2 Methodology

This section reviews the spatio-temporal skew-t process (STP) of Morris, Reich, Emeric Thibaud, and Cooley (2017), develops the AR(2) prior on the Gaussian-copula margins of its latent processes and the multi-resolution thin-plate spline (MRTS) augmentation of its regression mean, and assembles the full hierarchical model with the MCMC algorithm used for inference.

2.1 The spatio-temporal skew-t model

Let $Y_t(\mathbf{s})$ denote the response at spatial location $\mathbf{s} \in \mathcal{D} \subset \mathcal{R}^2$ on day $t \in \{1, \ldots, n_t\}$. Following Morris, Reich, Emeric Thibaud, and Cooley (2017), we model

$$Y_t(\mathbf{s}) \;=\; \mathbf{X}_t(\mathbf{s})^\top \boldsymbol{\beta} \;+\; \lambda\, \sigma_t(\mathbf{s})\, |z_t(\mathbf{s})| \;+\; \sigma_t(\mathbf{s})\, v_t(\mathbf{s}), \tag{1}$$

where $\mathbf{X}_t(\mathbf{s})$ is a p-vector of covariates, $\boldsymbol{\beta}$ is the corresponding regression coefficient vector, $\lambda \in \mathcal{R}$ is a scalar skewness parameter, and $v_t(\mathbf{s})$ is a standard Gaussian process with stationary isotropic Matérn correlation

$$\rho(h) \;=\; (1-\eta)\, I(h=0) \;+\; \eta\, \frac{1}{\Gamma(\nu)\, 2^{\nu-1}} \left(\sqrt{2\nu}\, \frac{h}{\varphi}\right)^{\nu} K_\nu\!\left(\sqrt{2\nu}\, \frac{h}{\varphi}\right), \tag{2}$$

where $\varphi > 0$ is the spatial range, $\nu > 0$ is the smoothness, $\eta \in [0, 1]$ is the proportion of variance attributed to the spatial component, K_ν is the modified Bessel function of the second kind, and $h = \|\mathbf{s}_1 - \mathbf{s}_2\|$ is the Euclidean distance between sites $\mathbf{s}_1$ and $\mathbf{s}_2$. Marginalizing over $|z_t(\mathbf{s})|$ and $\sigma_t(\mathbf{s})$, the finite-dimensional vector $\mathbf{Y}_t = [Y_t(\mathbf{s}_1), \ldots, Y_t(\mathbf{s}_n)]^\top$ follows an n-dimensional skew-t distribution; consequently (1) provides asymmetric and heavy-tailed margins together with asymptotic dependence between nearby sites.

Censoring. Because the EPA compliance criterion focuses on exceedances of a high level L, the model fits only the upper tail by censoring the data at a pre-specified threshold $T \leq L$. Writing the partially censored response $\widetilde{Y}_t(\mathbf{s}) = \max\{Y_t(\mathbf{s}), T\}$, inference is based on $\widetilde{\mathbf{Y}}_t = [\widetilde{Y}_t(\mathbf{s}_1), \ldots, \widetilde{Y}_t(\mathbf{s}_n)]^\top$, and the censored values below T are imputed inside the MCMC.

Random Partition. To remove the long-range asymptotic dependence that arises when a single scalar z and σ^2 are shared across the entire domain, the model uses a daily random partition. For each day t, let $\mathbf{w}_{t1}, \ldots, \mathbf{w}_{tK}$ denote K spatial knots in $\mathcal{D}$, and define the partition subregions

$$P_{tk} = \{\mathbf{s} \in \mathcal{D} : k = \arg\min_\ell \|\mathbf{s} - \mathbf{w}_{t\ell}\|\}, \qquad k = 1, \ldots, K. \tag{3}$$

For $\mathbf{s} \in P_{tk}$ we set $z_t(\mathbf{s}) = z_{tk}$ and $\sigma_t^2(\mathbf{s}) = \sigma_{tk}^2$, with $z_{tk} \overset{iid}{\sim} N(0, 1)$ and $\sigma_{tk}^2 \overset{iid}{\sim} \mathrm{IG}(a/2, b/2)$. Within each subregion the model retains the skew-t form, while sites in different partitions are not asymptotically dependent.

Temporal Dependence. Daily temporal dependence is introduced on the latent parameters that drive the tail, namely the knot locations $\mathbf{w}_{tk}$, the skewness latents z_{tk}, and the scale parameters σ_{tk}^2. To preserve the marginal skew-t distribution at every day, each parameter is first transformed to a standard normal scale via the probability integral transform (PIT):

$$w_{tki}^* = \Phi^{-1}\left[\frac{w_{tki} - \min(\mathbf{s}_i)}{\max(\mathbf{s}_i) - \min(\mathbf{s}_i)}\right], \quad i = 1, 2, \tag{4}$$

$$z_{tk}^* = \Phi^{-1}\{\mathrm{HN}(z_{tk})\}, \tag{5}$$

$$\sigma_{tk}^{2*} = \Phi^{-1}\{\mathrm{IG}(\sigma_{tk}^2)\}, \tag{6}$$

where Φ is the standard normal distribution function, HN is the half-normal distribution function, and IG is the inverse-gamma distribution function with parameters $(a/2, b/2)$. After transformation, each $\mathbf{w}^*_{tk}, z^*_{tk}, \sigma^{2*}_{tk}$ has a standard normal marginal distribution, and Morris, Reich, Emeric Thibaud, and Cooley (2017) place a stationary AR(1) time series on each of them:

$$\mathbf{w}^*_{tk} \mid \mathbf{w}^*_{t-1,k} \sim N_2\big(\phi^{(w)}\, \mathbf{w}^*_{t-1,k},\ (1-(\phi^{(w)})^2)\, \mathbf{I}_2\big)\,, \tag{7}$$

$$z^*_{tk} \mid z^*_{t-1,k} \sim N\big(\phi^{(z)}\, z^*_{t-1,k},\ 1-(\phi^{(z)})^2\big)\,, \tag{8}$$

$$\sigma^{2*}_{tk} \mid \sigma^{2*}_{t-1,k} \sim N\big(\phi^{(\sigma)}\, \sigma^{2*}_{t-1,k},\ 1-(\phi^{(\sigma)})^2\big)\,, \tag{9}$$

with $|\phi^{(w)}|, |\phi^{(z)}|, |\phi^{(\sigma)}| < 1$, and initial values $\mathbf{w}^*_{1k} \sim N_2(\mathbf{0}, \mathbf{I}_2)$, $z^*_{1k} \sim N(0,1)$, $\sigma^{2*}_{1k} \sim N(0,1)$. Transforming back through (4)–(6) gives $\mathbf{w}_{tk} \sim \mathrm{Unif}(\mathcal{D})$, $z_{tk} \sim \mathrm{HN}(0,1)$, and $\sigma^2_{tk} \sim \mathrm{IG}(a/2, b/2)$ at every t, so that the spatial skew-t structure of (1) is preserved at each day.

2.2 Extremal dependence

For exceedance prediction, what matters is how strongly the skew-t process couples extreme values at two sites, and how that coupling changes with their separation. One measure of this extremal dependence is the χ statistic (Coles, Heffernan, and J. Tawn, 1999), which for a stationary and isotropic spatial process at separation h is

$$\chi(h) = \lim_{c \to c^*} \Pr[Y(\mathbf{s}+h) > c \mid Y(\mathbf{s}) > c], \tag{10}$$

where c^* is the upper limit of the support of Y; for the skew-t distribution $c^* = \infty$. If $\chi(h) = 0$, then observations are asymptotically independent at distance h. For Gaussian processes, $\chi(h) = 0$ regardless of the distance h, so they are not suitable for modeling asymptotically dependent extremes. Unlike the Gaussian process, the skew-t process is asymptotically dependent. However, one problem with the spatial skew-t process is that $\lim_{h\to\infty} \chi(h) > 0$. This occurs because all observations, both near and far, share the same z and σ terms. Such long-range dependence is undesirable over large domains, where only local dependence is expected. Morris, Reich, Emeric Thibaud, and Cooley (2017) avoid this problem through a random partitioning mechanism, described in detail in Section 2.1.

2.3 Extension to AR(2) latent processes

In this section, we generalize the AR(1) prior (7)–(9) on the transformed latents to a stationary AR(2) prior.

The AR(2) prior. Writing $\{Y_t\}$ for any scalar component of $\mathbf{w}^*_{tk}$, or for z^*_{tk} or σ^{2*}_{tk} at a fixed knot k, the prior is the stationary AR(2) recursion

$$Y_t \;=\; \phi_1 \, Y_{t-1} \;+\; \phi_2 \, Y_{t-2} \;+\; \epsilon_t, \qquad \epsilon_t \overset{iid}{\sim} N(0, \sigma_\epsilon^2), \qquad t \geq 3, \tag{11}$$

with (ϕ_1, ϕ_2) confined to the stationarity triangle

$$|\phi_2| < 1, \qquad \phi_1 + \phi_2 < 1, \qquad \phi_2 - \phi_1 < 1. \tag{12}$$

Because the inverse PIT (4)–(6) returns the half-normal, inverse-gamma, and uniform target margins only when $\gamma_0 := \mathrm{Var}(Y_t) = 1$, the innovation variance is not free. Solving the Yule–Walker equations under $\gamma_0 = 1$ (Appendix A.2) fixes the standardized autocovariances

$$\gamma_1 \;=\; \frac{\phi_1}{1 - \phi_2}, \qquad \gamma_2 \;=\; \phi_1 \, \gamma_1 + \phi_2, \tag{13}$$

$$\sigma_\epsilon^2 \;=\; 1 - \phi_1 \, \gamma_1 - \phi_2 \, \gamma_2. \tag{14}$$

Setting $\phi_2 = 0$ gives $\gamma_1 = \phi_1$ and $\sigma_\epsilon^2 = 1 - \phi_1^2$, the AR(1) form of (7)–(9), so the AR(2) prior strictly nests the prior of Morris, Reich, Emeric Thibaud, and Cooley (2017).

Stationary initial pair. The recursion (11) needs two seeds, and independent $N(0,1)$ seeds would leave $\mathrm{Var}(Y_t) \neq 1$ over the first days and break the margin. We therefore draw the initial pair from the stationary bivariate Gaussian

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \;\sim\; N_2\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \gamma_1 \\ \gamma_1 & 1 \end{pmatrix} \right), \tag{15}$$

so that $\mathrm{Var}(Y_t) = 1$ holds from $t = 1$ onward.

Application to the three transformed parameters. Mirroring the AR(1) block (7)–(9), the AR(2) prior (11) with initial condition (15) replaces it component-wise:

$$\mathbf{w}^*_{tk} \mid \mathbf{w}^*_{t-1,k}, \mathbf{w}^*_{t-2,k} \sim N_2\Big(\phi_1^{(w)}\, \mathbf{w}^*_{t-1,k} + \phi_2^{(w)}\, \mathbf{w}^*_{t-2,k},\ \sigma^2_{\epsilon,w}\, \mathbf{I}_2\Big)\,, \tag{16}$$

$$z^*_{tk} \mid z^*_{t-1,k}, z^*_{t-2,k} \sim N\Big(\phi_1^{(z)}\, z^*_{t-1,k} + \phi_2^{(z)}\, z^*_{t-2,k},\ \sigma^2_{\epsilon,z}\Big)\,, \tag{17}$$

$$\sigma^{2*}_{tk} \mid \sigma^{2*}_{t-1,k}, \sigma^{2*}_{t-2,k} \sim N\Big(\phi_1^{(\sigma)}\, \sigma^{2*}_{t-1,k} + \phi_2^{(\sigma)}\, \sigma^{2*}_{t-2,k},\ \sigma^2_{\epsilon,\sigma}\Big)\,, \tag{18}$$

where each pair $(\phi_1^{(\cdot)}, \phi_2^{(\cdot)})$ lies in the stationarity triangle (12), the innovation variances follow from (13)–(14), and the initial pairs follow (15) component-wise. After the inverse PIT (4)–(6), the back-transformed $\mathbf{w}_{tk}, z_{tk}, \sigma^2_{tk}$ keep the same marginals as the AR(1) baseline, so the spatial skew-t model (1) is preserved at every t. The Gibbs-within-MH full conditional that updates each Y_t now carries three prior factors (self, lag-1, lag-2); its explicit form and the Metropolis–Hastings acceptance ratio are given in Appendix A.3.

2.4 Incorporating MRTS covariates

Tzeng and Huang (2018) proposed the multi-resolution thin-plate spline (MRTS) basis, whose functions are ordered from coarse to increasingly fine resolution. This ordering lets a model add spatial detail one resolution at a time; we define the basis below on a 2-dimensional spatial domain. Consider n distinct locations, $\mathbf{s}_1, \cdots, \mathbf{s}_n$, where $\mathbf{s}_i = (x_{i1}, x_{i2})^\top \in \mathcal{R}^2$ for $i = 1, \ldots, n$, and let

$$\mathbf{R} = \begin{pmatrix} 1 & x_{11} & x_{12} \\ \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} \end{pmatrix}. \tag{19}$$

Then the MRTS basis functions $f_k(\mathbf{s})$ at a location $\mathbf{s} = (x_1, x_2)^\top$ are defined as

$$f_k(\mathbf{s}) = \begin{cases} 1, & k = 1, \\ x_{k-1}, & k = 2, 3, \\ \mu_{k-3}^{-1}\big\{\boldsymbol{\psi}(\mathbf{s}) - \boldsymbol{\Psi}\mathbf{R}(\mathbf{R}^\top\mathbf{R})^{-1}\,\mathbf{r}\big\}^\top \mathbf{v}_{k-3}, & k = 4, \ldots, n, \end{cases} \tag{20}$$

where $\mathbf{r} = (1, x_1, x_2)^\top$, $\boldsymbol{\psi}(\mathbf{s}) = (\psi_1(\mathbf{s}), \ldots, \psi_n(\mathbf{s}))^\top$ with

$$\psi_i(\mathbf{s}) \;=\; \frac{1}{8\pi}\, \|\mathbf{s} - \mathbf{s}_i\|^2 \, \log\big(\|\mathbf{s} - \mathbf{s}_i\|\big), \qquad i = 1, \ldots, n, \tag{21}$$

and $\boldsymbol{\Psi}$ is the $n \times n$ matrix with (i, j)-th element $\psi_j(\mathbf{s}_i)$, $\mathbf{v}_k$ is the k-th column of $\mathbf{V}$, and $\mathbf{V}\,\mathrm{diag}(\mu_1, \ldots, \mu_n)\,\mathbf{V}^\top$ is the eigen-decomposition of $\mathbf{Q}\,\boldsymbol{\Psi}\,\mathbf{Q}$ with $\mu_1 \geq \cdots \geq \mu_n$ and $\mathbf{Q} = \mathbf{I} - \mathbf{R}(\mathbf{R}^\top\mathbf{R})^{-1}\mathbf{R}^\top$.

The MRTS basis enters the STP through the regression component. Fix the number of basis functions $K_{\text{MRTS}} \geq 1$ and let

$$\mathbf{f}(\mathbf{s}) \;=\; \big(f_1(\mathbf{s}),\, f_2(\mathbf{s}),\, \ldots,\, f_{K_{\text{MRTS}}}(\mathbf{s})\big)^\top \tag{22}$$

collect the first K_{MRTS} basis functions defined in (20). Then the day-t covariate vector $\mathbf{X}_t(\mathbf{s})$ in (1) is replaced by

$$\begin{aligned} \mathbf{X}_t^*(\mathbf{s}) &= \big[\mathbf{f}(\mathbf{s})^\top,\; \mathbf{X}_t(\mathbf{s})^\top\big]^\top \\ &= \big[f_1(\mathbf{s}),\; f_2(\mathbf{s}),\; \ldots,\; f_{K_{\text{MRTS}}}(\mathbf{s}),\; \mathrm{X}_1(\mathbf{s}, t),\; \ldots,\; \mathrm{X}_p(\mathbf{s}, t)\big]^\top, \end{aligned} \tag{23}$$

where $\mathrm{X}_j(\mathbf{s}, t)$ represents the j-th exogenous covariate of $\mathbf{X}_t(\mathbf{s})$. The model in (1) can then be expressed as

$$Y_t(\mathbf{s}) \;=\; \mathbf{X}_t^*(\mathbf{s})^\top \boldsymbol{\beta}^* \;+\; \lambda\,\sigma_t(\mathbf{s})\,|z_t(\mathbf{s})| \;+\; \sigma_t(\mathbf{s})\,v_t(\mathbf{s}), \tag{24}$$

which is identical to the STP model (1) on all components other than the mean, with $\boldsymbol{\beta}^* \in \mathcal{R}^{K_{\text{MRTS}}+p}$. Because the MRTS basis functions depend only on location, each site $\mathbf{s}$ shares the same ordered set of basis values $\mathbf{f}(\mathbf{s})$ at every day t. The number of basis functions K_{MRTS} is the practical tuning parameter. Section 3.4 compares $K_{\text{MRTS}} \in \{0, 3, 5, 8, 10, 12, 15, 20, 25\}$ across its simulation designs.

2.5 Hierarchical model

We specify the model as a three-stage Bayesian hierarchical model in the sense of Berliner (1996), following the presentation of hierarchical spatial models in Gelfand et al. (2010): a data stage conditional on the latent process, a process stage conditional on the parameters, and a parameter stage carrying the priors. Collect the day-t latent processes as $\mathcal{Z}_t = \{(\mathbf{w}_{tk},\, z_{tk},\, \sigma_{tk}^2) : k = 1, \ldots, K\}$ and the parameters as $\boldsymbol{\theta} = \{\boldsymbol{\beta}, \lambda, \varphi, \nu, \eta, a, b, \boldsymbol{\phi}\}$, where

ϕ collects the AR coefficients of the three latent processes. The number of knots K is not treated as random; it indexes candidate models selected by held-out Brier score (Section 3.2).

Stage 1: data model. The observations are the censored responses,

$$\widetilde{Y}_t(\mathbf{s}_i) \;=\; \max\{Y_t(\mathbf{s}_i),\, T\}, \qquad i=1,\dots,n, \quad t=1,\dots,n_t, \tag{25}$$

so that values above T are observed exactly, while values below T are known only to lie below T and are imputed within the MCMC.

Stage 2: process model. The first substage is the conditional Gaussian field: independently across days,

$$\mathbf{Y}_t \mid \mathcal{Z}_t, \boldsymbol{\theta} \;\sim\; N_n\Big(\mathbf{X}_t\boldsymbol{\beta} \,+\, \lambda\,\mathbf{D}_t\,|\mathbf{z}_t|,\;\; \mathbf{D}_t\,\mathbf{R}(\varphi,\nu,\eta)\,\mathbf{D}_t\Big), \tag{26}$$

where $\mathbf{D}_t = \operatorname{diag}\{\sigma_t(\mathbf{s}_1),\dots,\sigma_t(\mathbf{s}_n)\}$, $|\mathbf{z}_t| = (|z_t(\mathbf{s}_1)|,\dots,|z_t(\mathbf{s}_n)|)^\top$, and $\mathbf{R}(\varphi,\nu,\eta)$ has entries $\rho(\|\mathbf{s}_i-\mathbf{s}_j\|)$ from (2). Under the MRTS extension, $\mathbf{X}_t\boldsymbol{\beta}$ is replaced by $\mathbf{X}_t^*\boldsymbol{\beta}^*$ of (23).

The second substage maps the subregion latents to the sites through the partition: for $\mathbf{s}\in P_{tk}$,

$$z_t(\mathbf{s}) \;=\; z_{tk}, \qquad \sigma_t^2(\mathbf{s}) \;=\; \sigma_{tk}^2, \qquad P_{tk} \;=\; \{\mathbf{s}\in\mathcal{D} : k = \arg\min_\ell \|\mathbf{s}-\mathbf{w}_{t\ell}\|\}. \tag{27}$$

The third substage gives the latent temporal dynamics on the transformed scale of (4)–(6), independently across knots $k=1,\dots,K$: for $t\ge 3$,

$$\mathbf{w}_{tk}^* \mid \mathbf{w}_{t-1,k}^*,\, \mathbf{w}_{t-2,k}^*,\, \boldsymbol{\phi}^{(w)} \sim N_2\big(\phi_1^{(w)}\mathbf{w}_{t-1,k}^* + \phi_2^{(w)}\mathbf{w}_{t-2,k}^*,\; \sigma_{\epsilon,w}^2\,\mathbf{I}_2\big), \tag{28}$$
$$z_{tk}^* \mid z_{t-1,k}^*,\, z_{t-2,k}^*,\, \boldsymbol{\phi}^{(z)} \sim N\big(\phi_1^{(z)} z_{t-1,k}^* + \phi_2^{(z)} z_{t-2,k}^*,\; \sigma_{\epsilon,z}^2\big), \tag{29}$$
$$\sigma_{tk}^{2*} \mid \sigma_{t-1,k}^{2*},\, \sigma_{t-2,k}^{2*},\, \boldsymbol{\phi}^{(\sigma)} \sim N\big(\phi_1^{(\sigma)}\sigma_{t-1,k}^{2*} + \phi_2^{(\sigma)}\sigma_{t-2,k}^{2*},\; \sigma_{\epsilon,\sigma}^2\big), \tag{30}$$

with each innovation variance fixed by the Yule–Walker standardization (13)–(14) and each initial pair drawn from the stationary bivariate Gaussian (15). The AR(1) baseline of Morris, Reich, Emeric Thibaud, and Cooley (2017) is the special case $\phi_2^{(\cdot)}=0$, which reduces (28)–(30) to (7)–(9). The inverse transforms (4)–(6) return the marginals $\mathbf{w}_{tk}\sim\operatorname{Unif}(\mathcal{D})$, $z_{tk}\sim \operatorname{HN}(0,1)$, and $\sigma_{tk}^2\sim\operatorname{IG}(a/2,\,b/2)$ at every t.

Stage 3: parameter model. All parameters, and only parameters, receive priors at this stage:

$$
\begin{aligned}
&\boldsymbol{\beta} \sim N_p(\mathbf{0},\, \sigma_\beta^2 \mathbf{I}), && \lambda \sim N(0,\, \sigma_\lambda^2), && \eta \sim \text{Unif}(0, 1),\\
&\varphi \sim \text{Unif}(0,\, \varphi_{\text{max}}), && a/2 \sim \text{Unif}\{0.05,\, 0.10,\, \ldots,\, 5\}, && b/2 \sim \text{Gamma}(1,\, 1),\\
&\log \nu \sim N(-1.2,\, 1),\ \nu \le 10, && (\phi_1^{(\cdot)},\, \phi_2^{(\cdot)}) \sim \ \text{Unif}\big(\text{stationarity triangle (12)}\big). &&
\end{aligned} \tag{31}
$$

Here $\text{Unif}\{\cdot\}$ denotes the discrete uniform distribution on the listed grid, and the hyperparameters are $\sigma_\beta = \sigma_\lambda = 20$, with the range bound $\varphi_{\text{max}} = 15$ in the simulation study and $\varphi_{\text{max}} = 5$ in the ozone study, matched to the extent of each spatial domain. The smoothness prior is lognormal with median $e^{-1.2} \approx 0.30$, truncated above at 10, so ν is estimated in every fit rather than fixed at the exponential case $\nu = 0.5$ used to generate the simulation designs.

Writing $[\,\cdot\,]$ for a generic distribution, the three stages multiply to the joint model, and by Bayes' Rule the posterior is proportional to that product,

$$
\big[\mathbf{Y}_{\text{cens}},\, \mathcal{Z},\, \boldsymbol{\theta} \mid \widetilde{\mathbf{Y}}\big] \propto \prod_{t=1}^{n_t} \big[\widetilde{\mathbf{Y}}_t \mid \mathbf{Y}_t\big] \big[\mathbf{Y}_t \mid \mathcal{Z}_t,\, \boldsymbol{\theta}\big] \big[\mathcal{Z}_{1:n_t} \mid \boldsymbol{\theta}\big] \big[\boldsymbol{\theta}\big], \tag{32}
$$

sampled by Gibbs-within-MH updates: each factor of (32) is one of the stages above, and the full conditionals are read off stage by stage; the AR(2) full conditional and its Metropolis–Hastings acceptance ratio are given in Appendix A.3. Predictions at unmonitored sites follow from the conditional Gaussian structure of (26) (Bayesian kriging).

3 Simulation study

This simulation study examines whether, and under which data-generating settings, the AR(2) and MRTS extensions improve held-out exceedance prediction. Section 3.1 gives the design shared by all experiments and Section 3.2 the validation protocols, including the uncertainty attached to every score comparison. Section 3.3 evaluates the AR(2) extension under autoregressive, highly persistent, and long-memory temporal dynamics; Section 3.4 evaluates the MRTS extension under second-moment and first-moment spatial non-stationarity. Throughout, both extensions are assessed as candidate options, benchmarked against Gaussian fits and against matched models without each extension.

3.1 Common simulation design

All simulation designs generate data from the hierarchical model of Section 2.5 on $n_s = 144$ sites, drawn uniformly on $[0, 10] \times [0, 10]$, over $n_t = 50$ days ($n_t = 200$ in the time-block forecast evaluation of Section 3.3); the single exception is the pair of nonlinear-mean designs of Section 3.4, which replace the skew-t error process with a Gaussian one. The Matérn correlation (2) is generated at $\eta = 0.9$, $\nu = 1/2$, and $\varphi = 1$ in every design. In the data generation the regression mean is held constant, $\mathbf{X}^\top \boldsymbol{\beta} = 10$ (Section 3.4 adds a mean-zero non-stationary random effect to one design, which leaves the regression mean at 10, and a fixed nonlinear mean surface to its two Gaussian-error designs), while every fitted model includes an intercept and a first-order spatial trend. For each design we simulate 10 data sets, each split into 100 training sites and 44 withheld test sites, and run every MCMC chain for $20{,}000$ iterations with a burn-in period of $10{,}000$ iterations. Throughout, $q(0.80)$ denotes the 80th sample quantile of the data, and the five baseline analysis models are

1. Gaussian marginal, $K = 1$ knot;
2. Skew-t marginal, $K = 1$ knot, no thresholding;
3. Symmetric-t marginal, $K = 1$ knot, threshold $T = q(0.80)$;
4. Skew-t marginal, $K = 5$ knots, no thresholding;
5. Symmetric-t marginal, $K = 5$ knots, threshold $T = q(0.80)$.

Censoring is paired with the symmetric-t marginal because the skewness parameter is only weakly identified once the lower tail is censored, following Morris, Reich, Emeric Thibaud, and Cooley (2017).

3.2 Validation protocols and uncertainty

Spatial hold-out validation. Models are compared on held-out sites, with 100 training sites fitting each setting and 44 withheld sites scoring its predictions on each replicate. Because the target is exceedance of a high level, we compare models by the Brier score (Gneiting and Raftery, 2007). For an exceedance level L, the Brier score is

$$\mathrm{BS}(L) = \left\{ I(y > L) - \widehat{P}(Y > L) \right\}^2, \tag{33}$$

where y is the held-out observation, $I(y > L)$ is the exceedance indicator, and $\widehat{P}(Y > L)$ is the posterior predictive probability of exceeding L. We average the Brier scores over all test sites and days; lower values indicate better exceedance prediction, and L ranges over the sample quantiles reported with each figure.

Time-block forecast validation. The site-splitting scheme above scores interpolation. When a withheld site is predicted, every day of the record is observed at the training sites. The time-block forecast evaluation of Section 3.3 scores temporal extrapolation instead, with all 144 sites observed for days 1–50 and predictions evaluated at the same sites on the following 15 days. For each forecast lead h the Brier score (33) is computed with L set to the empirical 0.95 quantile of the validation block (the same threshold for every model), averaged over sites, and then over data sets; models are compared pairwise within each data set. Short-record fits whose skewness parameter and intercept fail to separate, or whose predictive spread is unbounded, are screened before scoring.

Uncertainty of score comparisons. A mean Brier score is an estimate, and two such estimates computed on the same held-out sites are strongly positively correlated, since a site that is hard to predict is hard for every model. We therefore attach uncertainty to the *paired difference* of scores. For a matched pair of models A and B and an exceedance level L, the estimand is

$$\Delta(L) \;=\; \mathrm{E}\big[\mathrm{BS}_A(L)\big] \;-\; \mathrm{E}\big[\mathrm{BS}_B(L)\big], \tag{34}$$

estimated by forming the difference site by site on the shared held-out set, so that the common hard-site variation cancels, and assessing its sampling variability over the 10 simulated data sets. Each comparison tests $H_0\colon \Delta(L) = 0$ against the two-sided alternative, summarized by a 95% paired-t confidence interval and a Wilcoxon signed-rank test, which does not assume normal differences; with 10 replicates the Wilcoxon p-values move on a coarse grid whose smallest attainable two-sided value is $2^{-9} \approx 0.002$. Each pair varies only the factor under test, with the analysis knot count matched to the generating design. In the tables that follow, a star marks an interval that excludes zero; among the dozens of paired comparisons reported in this thesis, an isolated borderline star is within what chance alone would produce, and an interval that covers zero means the data fail to reject H_0, not that the two models have been shown equal. In the ozone study of Section 4 there is a single data set and hence no replicate layer, so the resampling unit is instead the validation site, as described there.

3.3 The AR(2) experiment

Design. To test the AR(2) analysis model against data with a known autoregressive structure, the six generating designs are skew-t ($\lambda = 3$) with temporal dependence imposed on the latent σ^2, $\mathbf{w}$, and z processes. The first five evolve through a fixed AR(2) recursion (16)–(18); the sixth replaces it with a long-memory ARFIMA$(0, d, 0)$ process:

1. $K = 1$ knot, strong dependence $(\phi_1, \phi_2) = (0.80, -0.35)$;
2. $K = 1$ knot, weak dependence $(\phi_1, \phi_2) = (0.12, -0.05)$;
3. $K = 5$ knots, strong dependence $(\phi_1, \phi_2) = (0.80, -0.35)$;
4. $K = 5$ knots, weak dependence $(\phi_1, \phi_2) = (0.12, -0.05)$;
5. $K = 1$ knot, persistent AR(2) $(\phi_1, \phi_2) = (0.15, 0.80)$;
6. $K = 1$ knot, long-memory ARFIMA$(0, d, 0)$, $d = 0.45$ (Davies and Harte, 1987).

All three AR(2) coefficient pairs lie inside the stationarity triangle (12). Each of the first four designs is fit by nine analysis models, namely the five baselines and the skew-t models with AR(1) and AR(2) coefficients at $K = 1$ and $K = 5$ knots.

The persistent and long-memory designs (5 and 6) are evaluated under both protocols of Section 3.2. The spatial hold-out evaluation fits the Gaussian model and the $K = 1$ skew-t models with AR(1) and AR(2) structure; the time-block forecast evaluation fits the $K = 1$ skew-t variants with the latent processes i.i.d. in time, with AR(1) structure, and with AR(2) structure. As a short-memory control, the fixed-AR(2) design with $(\phi_1, \phi_2) = (0.80, -0.35)$, whose autocorrelation decays within a few days, is run under the same forecast protocol with the i.i.d. and AR(2) variants.

Results. In the spatial hold-out evaluation, predictive ranking is governed by the marginal and the spatial partition. The methods whose marginal matches the generating truth give the largest improvement over the Gaussian model, a mismatched marginal can perform worse than Gaussian in the far tail, and the preferred knot count tracks the generating truth (Figure 1(a)–(d)). The temporal structure barely moves these scores. The AR(2) and AR(1) curves are essentially indistinguishable from their non-temporal counterparts at every quantile and in every setting, and the paired intervals of Table 2 confirm the visual impression; 32 of the 33 intervals cover zero, the single starred exception ($K = 5$, $(0.80, -0.35)$, at the 0.90 quantile)

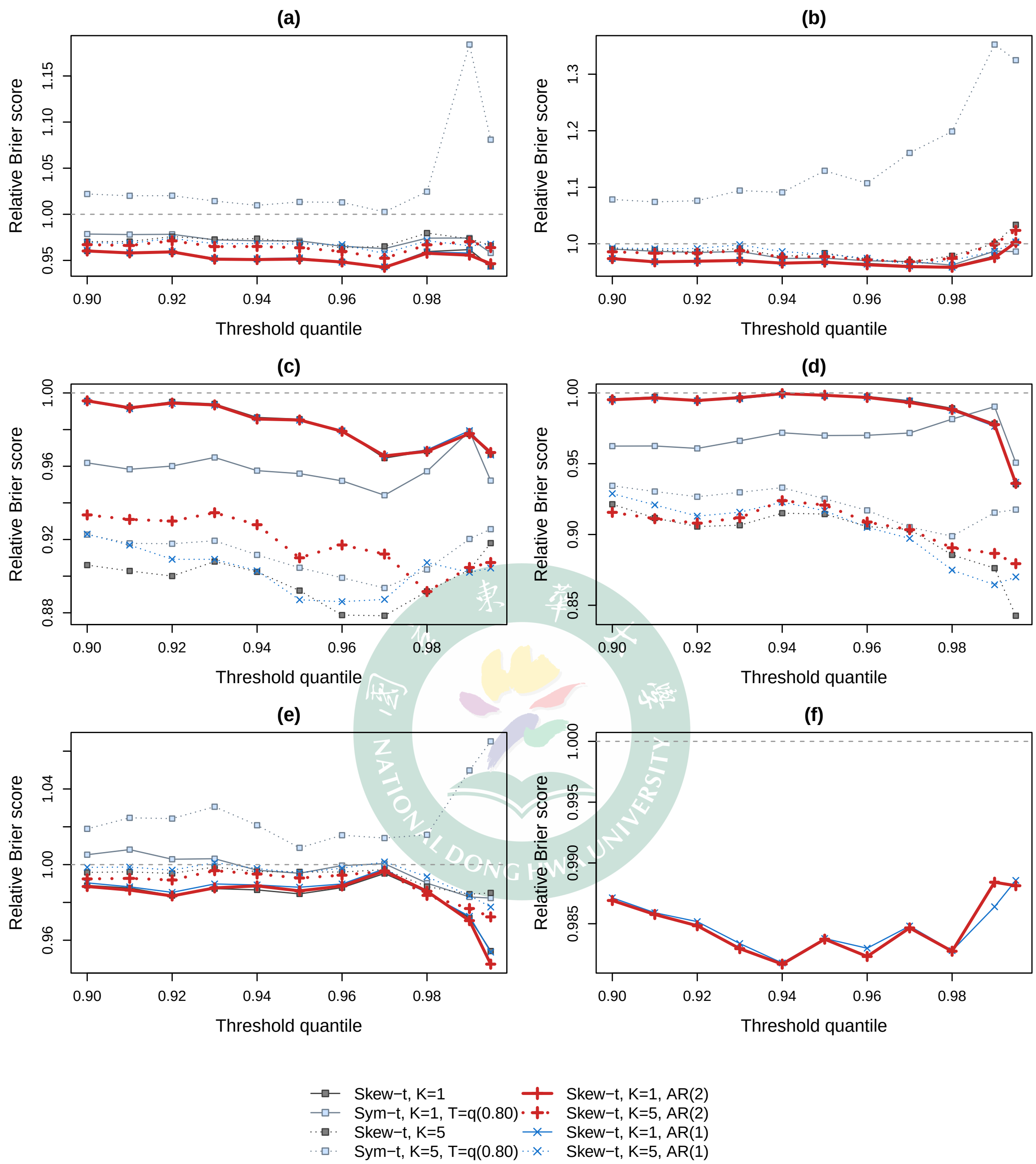


Figure 1: Spatial hold-out evaluation: relative Brier score versus the Gaussian model by threshold quantile (Gaussian is the reference $\equiv 1$, shown as the dashed line; values below it are better). All data-generating settings are skew-t with $\lambda = 3$: (a) $K = 1$, AR(2) $(\phi_1, \phi_2) = (0.80, -0.35)$; (b) $K = 1$, AR(2) $(0.12, -0.05)$; (c) $K = 5$, AR(2) $(0.80, -0.35)$; (d) $K = 5$, AR(2) $(0.12, -0.05)$; (e) $K = 1$, AR(2) $(0.15, 0.80)$; (f) $K = 1$, long memory, ARFIMA$(0, d, 0)$ with $d = 0.45$. Panel (f) draws only the one-knot AR(1) and AR(2) fits, the only models fit under the long-memory design.

favors the *non-temporal* baseline and sits at the edge of significance, within what chance alone produces among this many comparisons, and the two-sided Wilcoxon p-values, ranging from 0.05 to 1.00, agree.

Table 2: Spatial hold-out evaluation of the AR(2) experiment: paired Brier differences over the 10 data sets, $\Delta = \mathrm{BS}_A - \mathrm{BS}_B$ with 95% paired-t confidence intervals. A star marks an interval that excludes zero.

		$\Delta \times 10^3$ [95% CI]	
Generating design	L quantile	AR(2) − i.i.d.	AR(2) − AR(1)
K=1, $(0.80, -0.35)$	0.90	$+0.00$ $[-0.03, +0.04]$	-0.03 $[-0.06, +0.00]$
	0.95	-0.01 $[-0.03, +0.01]$	-0.02 $[-0.06, +0.02]$
	0.98	-0.01 $[-0.04, +0.01]$	-0.01 $[-0.03, +0.01]$
K=1, $(0.12, -0.05)$	0.90	$+0.00$ $[-0.02, +0.03]$	-0.01 $[-0.04, +0.02]$
	0.95	-0.00 $[-0.03, +0.02]$	-0.01 $[-0.03, +0.01]$
	0.98	-0.01 $[-0.03, +0.01]$	$+0.00$ $[-0.01, +0.01]$
K=5, $(0.80, -0.35)$	0.90	$+0.93^*$ $[+0.12, +1.75]$	$+0.36$ $[-0.51, +1.23]$
	0.95	$+0.32$ $[-0.33, +0.97]$	$+0.41$ $[-0.12, +0.94]$
	0.98	-0.00 $[-0.33, +0.32]$	-0.14 $[-0.33, +0.05]$
K=5, $(0.12, -0.05)$	0.90	-0.21 $[-1.16, +0.74]$	-0.48 $[-1.16, +0.20]$
	0.95	$+0.13$ $[-0.27, +0.52]$	$+0.08$ $[-0.34, +0.50]$
	0.98	$+0.04$ $[-0.26, +0.35]$	$+0.13$ $[-0.37, +0.63]$
K=1, $(0.15, 0.80)$	0.90	-0.02 $[-0.07, +0.03]$	-0.07 $[-0.23, +0.09]$
	0.95	$+0.04$ $[-0.00, +0.08]$	-0.05 $[-0.19, +0.10]$
	0.98	$+0.01$ $[-0.02, +0.03]$	$+0.01$ $[-0.02, +0.04]$
ARFIMA, $d = 0.45$	0.90	–	-0.01 $[-0.04, +0.02]$
	0.95	–	-0.00 $[-0.03, +0.03]$
	0.98	–	-0.00 $[-0.04, +0.03]$

Coefficient recovery separates the two possible readings of this insensitivity (Table 3). In the four fixed designs the temporal signal is only weakly identified; the scale process alone is recovered with any fidelity, under strong dependence at a matched one-knot fit, and recovery deteriorates as the knot count grows and the dependence weakens. In the persistent and long-memory designs (5 and 6) the signal is identified; the matched fit recovers the generating dynamics closely, in particular the dominant second lag that an AR(1) can only compress into a single lag, and even under long memory the AR(2) fit picks up a clearly positive second lag. Yet the hold-out scores of these two designs repeat the null pattern in sharper form (Figure 1(e)–(f); Table 2, lower block). Identified temporal parameters with unmoved scores point to the task rather than the model, exactly the interpolation dominance that the protocol distinction of Section 3.2 anticipates.

Table 3: Posterior mean (SD) of the recovered AR(2) coefficients across the generating designs of the AR(2) experiment: the four fixed-AR(2) settings (upper block), the persistent AR(2) design $(0.15, 0.80)$, and the long-memory ARFIMA design $d = 0.45$ (lower block). Gen K is the data-generating knot count and Fit K the analysis-model knot count; $\hat{\phi}^{(\sigma)}$, $\hat{\phi}^{(z)}$, $\hat{\phi}^{(w)}$ are the estimated AR(2) coefficients of the scale, skewness, and knot-location processes ((16)–(18)).

Settings		Fit	$\hat{\phi}^{(\sigma)}$		$\hat{\phi}^{(z)}$		$\hat{\phi}^{(w)}$	
(ϕ_1, ϕ_2)	K	K	$\hat{\phi}_1$	$\hat{\phi}_2$	$\hat{\phi}_1$	$\hat{\phi}_2$	$\hat{\phi}_1$	$\hat{\phi}_2$
$(0.80, -0.35)$	1	1	0.803(0.140)	-0.363(0.137)	0.626(0.114)	-0.225(0.157)	-	-
$(0.80, -0.35)$	1	5	0.871(0.142)	-0.438(0.164)	0.472(0.284)	-0.270(0.162)	-0.129(0.377)	0.062(0.257)
$(0.12, -0.05)$	1	1	0.083(0.122)	-0.019(0.150)	0.186(0.137)	0.084(0.226)	-	-
$(0.12, -0.05)$	1	5	0.095(0.141)	-0.050(0.220)	0.050(0.088)	0.051(0.142)	0.002(0.215)	0.179(0.259)
$(0.80, -0.35)$	5	1	0.512(0.190)	0.043(0.245)	0.434(0.185)	-0.108(0.128)	-	-
$(0.80, -0.35)$	5	5	0.169(0.115)	-0.023(0.118)	0.074(0.093)	-0.006(0.090)	-0.019(0.206)	0.094(0.135)
$(0.12, -0.05)$	5	1	0.285(0.228)	0.161(0.164)	0.163(0.122)	-0.063(0.122)	-	-
$(0.12, -0.05)$	5	5	-0.001(0.072)	-0.017(0.137)	0.010(0.075)	-0.010(0.067)	-0.017(0.193)	0.115(0.190)
$(0.15, 0.80)$	1	1	0.112(0.132)	0.742(0.117)	0.175(0.152)	0.728(0.112)	-	-
ARFIMA, d=0.45	1	1	0.502(0.237)	0.200(0.239)	0.365(0.249)	0.156(0.169)	-	-

The time-block forecast evaluation addresses the question the spatial split cannot answer, namely whether the fitted temporal structure improves genuine forward-in-time prediction (Figure 2, Table 4). Under long memory the answer is affirmative. The AR(2) model attains the lowest Brier score at almost every forecast lead, improving on the i.i.d.-in-time fit over both the short-lead window and the full horizon, with the AR(1) model falling in between, and this separation is far larger than any the spatial evaluation produced. Under the persistent design the per-lead profile is noisier, since each lead contributes a single binary exceedance event per site and day, and the three models track one another over the early leads; pooled over the full horizon the AR(2) fit is nevertheless the best of the three. Under the short-memory control $(0.80, -0.35)$ the two fits tie at both windows, the expected outcome once the autocorrelation has decayed within a few days of the forecast origin. None of the paired Brier differences reaches statistical significance. The largest gain, that of AR(2) over the i.i.d. fit under long memory at leads 1–5, is -0.0068 with a 95% paired-t interval of $(-0.033, 0.020)$, and the Wilcoxon p-values for the ten comparisons of Table 4 range from 0.31 to 1.00, so these results should be read as directional. The gains are modest, but they appear exactly where the

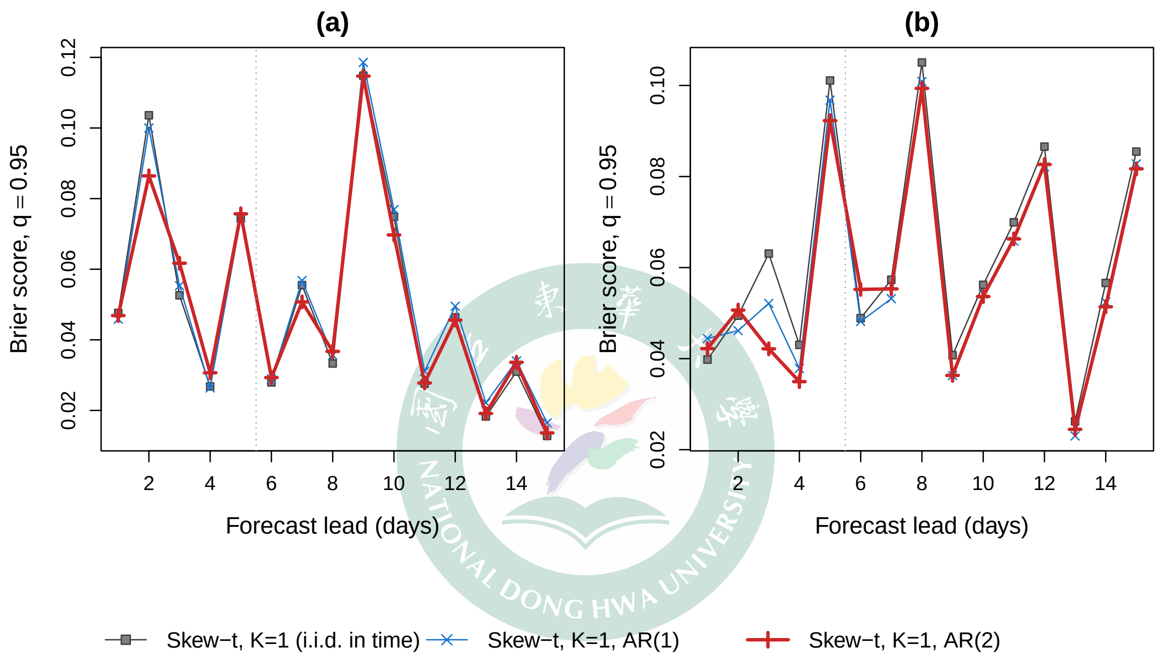


Figure 2: Time-block forecast evaluation of the persistence experiment: mean Brier score at the 0.95 threshold quantile by forecast lead (10 data sets; lower is better; the dotted vertical line marks the lead 1–5 window). Data-generating settings: (a) AR(2), $(\phi_1, \phi_2) = (0.15, 0.80)$; (b) long memory, ARFIMA$(0, d, 0)$ with $d = 0.45$.

structure of the task predicts them, namely in temporal extrapolation under strong persistence, and they are largest where the second lag improves the finite-order approximation.

Table 4: Time-block forecast evaluation: mean Brier score at the 0.95 threshold quantile by lead window (10 data sets; lower is better). The fitted models are the skew-t $K = 1$ variants with i.i.d., AR(1), and AR(2) latent temporal structure; the short-memory design $(0.80, -0.35)$ fit only the i.i.d. and AR(2) variants, marked '–'. The difference columns carry 95% paired-t confidence intervals over the 10 data sets.

		Mean Brier score			Δ [95% CI]	
Generating design	Leads	i.i.d.	AR(1)	AR(2)	AR(2)−i.i.d.	AR(2)−AR(1)
AR(2) $(0.80, -0.35)$	1–5	0.0526	–	0.0530	$+0.0004$ $[-0.0041,\ +0.0049]$	–
AR(2) $(0.80, -0.35)$	1–15	0.0597	–	0.0599	$+0.0002$ $[-0.0030,\ +0.0033]$	–
AR(2) $(0.15, 0.80)$	1–5	0.0610	0.0604	0.0603	-0.0007 $[-0.0077,\ +0.0062]$	-0.0001 $[-0.0066,\ +0.0064]$
AR(2) $(0.15, 0.80)$	1–15	0.0498	0.0515	0.0495	-0.0003 $[-0.0044,\ +0.0037]$	-0.0020 $[-0.0064,\ +0.0024]$
ARFIMA, $d = 0.45$	1–5	0.0593	0.0555	0.0524	-0.0068 $[-0.0332,\ +0.0196]$	-0.0030 $[-0.0163,\ +0.0103]$
ARFIMA, $d = 0.45$	1–15	0.0620	0.0584	0.0579	-0.0041 $[-0.0271,\ +0.0190]$	-0.0005 $[-0.0142,\ +0.0132]$

3.4 The MRTS experiment

Design. This experiment enriches the regression component with the MRTS basis functions $f_1(\mathbf{s}), \ldots, f_{K_{\text{MRTS}}}(\mathbf{s})$ of Section 2.4. The four generating designs place the non-stationarity first where the mean basis cannot reach it (designs 1–2) and then where it can (designs 3–4):

1. Skew-t , $K = 5$ knots, $\lambda = 3$, with the constant mean $\mathbf{X}^\top \boldsymbol{\beta} = 10$;

2. Skew-t , $K = 1$ knot, $\lambda = 3$, with an additive mean-zero non-stationary spatial random effect $u_t(\mathbf{s})$, redrawn daily from two fixed cosine fields (Appendix A.4), so the non-stationarity lies in the second moment while the marginal mean stays at 10;

3. Gaussian, arc front: the nonlinear-mean model (35) with $\tilde{g}(\mathbf{s}) = \tanh\{\|\mathbf{s} - (2, 2)^\top\| - 4.5\}$, a smoothed circular step;

4. Gaussian, transform mix: the nonlinear-mean model (35) with $\tilde{g}(\mathbf{s}) = \text{std}\{\log(1 + s_1)\} + \text{std}\{e^{-s_1/2}\} + \text{std}\{s_1/(1 + s_2)\}$, with $\text{std}\{\cdot\}$ standardizing each term to unit spatial standard deviation, whose curvature concentrates near the domain boundary, where the uniform site draw is sparsest.

The nonlinear-mean designs replace the skew-t generator with Gaussian errors,

$$Y_t(\mathbf{s}) \;=\; 10 \;+\; g(\mathbf{s}) \;+\; \sigma_g\, \varepsilon_t(\mathbf{s}), \qquad \sigma_g = 4, \tag{35}$$

where the errors ε_t are standard Gaussian processes with the Matérn correlation (2) at the generating values of Section 3.1, drawn independently across days; every t-family analysis model is therefore misspecified in its error law. Each raw surface $\tilde{g}$ is orthogonalized against the baseline design $[1, s_1, s_2]$ and standardized to a spatial standard deviation of 11, so the $K_{\text{MRTS}} = 0$ design spans none of the resulting g.

Each of the five baseline models is fit across a grid of MRTS basis sizes, $K_{\text{MRTS}} \in \{0, 3, 5, 8, 10, 12, 15, 20, 25\}$, where $K_{\text{MRTS}} = 0$ recovers the no-MRTS baseline; on both nonlinear-mean designs the grid is extended by $\{30, 40, 50\}$, since the finer surfaces are resolved only by a fine basis. All four designs are run with 10 data sets.

Results. On the two skew-t -core designs, augmenting the regression component with MRTS basis functions does not deliver a stable improvement in upper-tail scores, and enlarging the basis does not lower the average Brier score on either setting (Figure 3). On the five-knot setting every curve is essentially flat in K_{MRTS}; on the additive non-stationary random-effect process the curves are likewise flat, with over-specified models destabilizing at large K_{MRTS}. As in the AR(2) experiment, the generating knot count dominates the ranking. The second setting is the decisive one. The non-stationarity lives in the second moment (a daily-redrawn random effect u_t), whereas the MRTS basis enters only the fixed-effect mean. Because u_t has zero time-average, the time-pooled coefficient $\boldsymbol{\beta}^*$ has no first-moment structure to recover, so the Brier- and recovery-versus-K curves stay flat by construction rather than through absence of signal, since u_t contributes substantial (spatially non-stationary) variance. The MRTS mean term therefore yields no reliable tail-score gain. A fixed-effect mean cannot represent a non-stationary second-moment feature, and so, in these designs, the spatial mean is not the binding constraint on upper-tail prediction.

The two nonlinear-mean designs address the complementary case (Figure 4; Table 5). On the arc-front design the Brier score finally responds to the mean basis, but only once the basis is fine enough to resolve the front. The profiles stay flat for a coarse basis, deteriorate slightly at intermediate sizes, and improve only at the finer bases, with the thresholded models gaining more than the unthresholded ones and the gains larger at the higher exceedance level. The intermediate-basis dip is the signature of a too-coarse basis. A partially fitted mean sharpens the predictive distribution while leaving a residual surface error that extrapolates to the held-

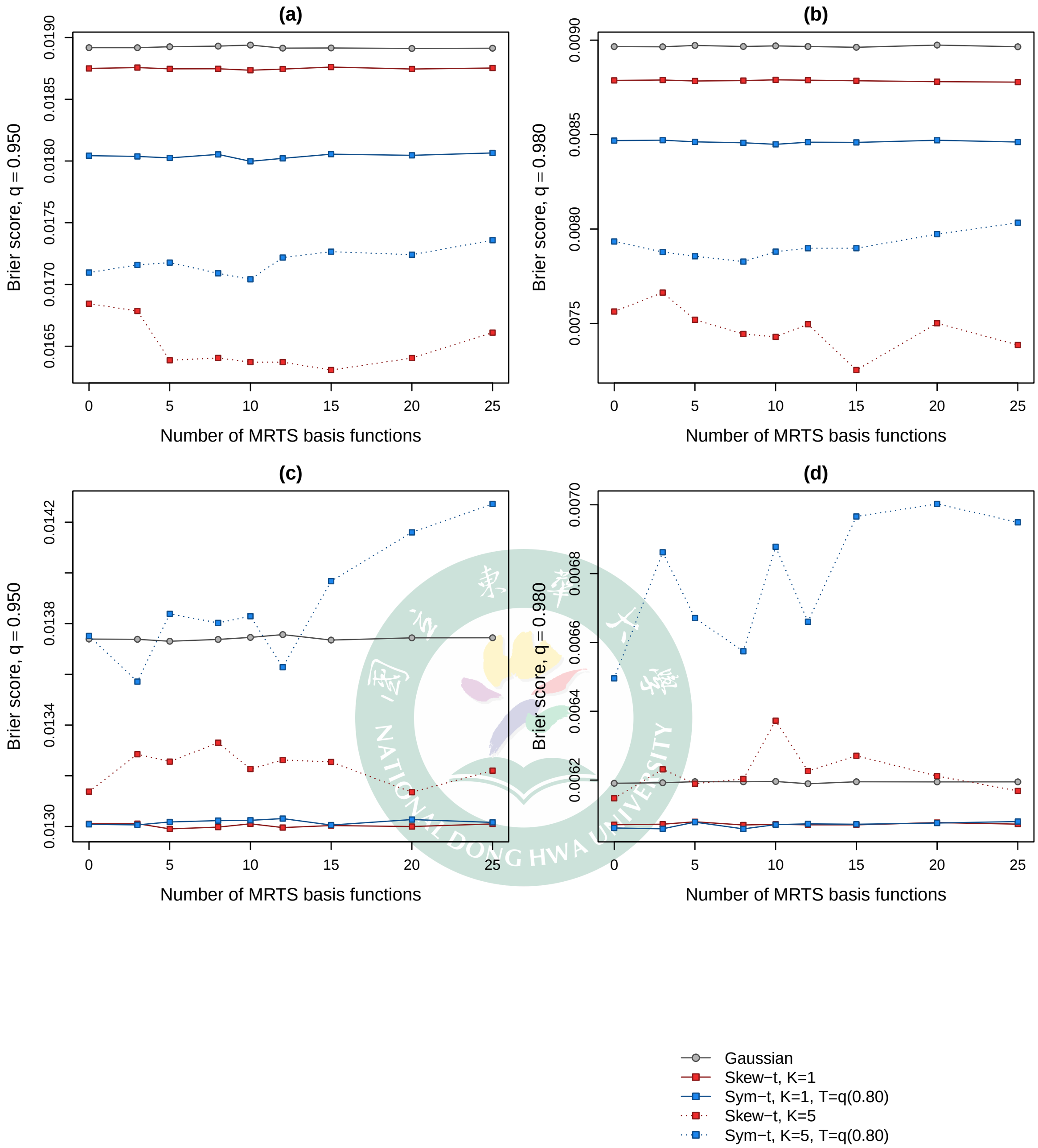


Figure 3: Brier score versus the number of MRTS basis functions K_{MRTS} ($K_{\text{MRTS}} = 0$ is the no-MRTS baseline; lower is better). Data-generating processes: (a,b) skew-t ($K = 5$, $\lambda = 3$); (c,d) skew-t ($K = 1$, $\lambda = 3$) with an additive non-stationary spatial random effect. Exceedance level $q = 0.95$ in (a,c) and $q = 0.98$ in (b,d).

out sites, and the net effect on the Brier score is negative until the basis actually resolves the front.

The transform-mix design shows the same mechanism operating from a much worse starting point. The mismatched five-knot partition, compounded for the thresholded variant by censoring, leaves the no-basis baselines of the five-knot models far above those of the one-knot models, and the mean basis recovers most of that loss. Every model improves at the 0.95 threshold quantile, and at its best basis size each converges to nearly the same score. The improvement is not uniform across levels, however. At the rarer 0.98 quantile it is carried entirely by the five-knot models, whereas the one-knot models stay flat or worsen. This is the sharpness–extrapolation signature again, in that only the models whose no-basis fit is badly mis-specified, here the five-knot partition, carry enough headroom for the mean basis to convert into a gain at the rarest level. Both surfaces were chosen, by an oracle calculation performed before any fitting, to leave genuine Brier headroom. Replacing the fitted mean by the true surface g under the true Gaussian error law leaves substantial headroom on both. The realized gains convert only a small part of that headroom, which is consistent with the sharpness–extrapolation account above. The Brier score improves only where the residual basis error is small relative to the predictive spread.

Table 5: Nonlinear-mean designs: Brier score at the 0.95 threshold quantile relative to the same model's own $K_{\text{MRTS}} = 0$ fit (lower is better). Best is the smallest ratio over $K_{\text{MRTS}} > 0$, with the minimizing basis size in parentheses; BS_0 is the absolute Brier score at $K_{\text{MRTS}} = 0$; thr. denotes the POT threshold $T = q(0.80)$. Both designs use 10 data sets on the full K_{MRTS} grid; the $K = 30, 40, 50$ columns are shown for comparability, and Best ranges over the whole grid.

Generating design	Analysis model	K=30	K=40	K=50	Best (K)	BS_0
Arc front	Gaussian	0.970	0.967	0.966	0.966 (50)	0.0232
Arc front	Skew-t , $K = 1$	0.969	0.967	0.967	0.966 (25)	0.0233
Arc front	Symmetric-t, $K = 1$, thr.	0.943	0.949	0.949	0.941 (25)	0.0240
Arc front	Skew-t , $K = 5$	0.960	0.956	0.956	0.956 (50)	0.0235
Arc front	Symmetric-t, $K = 5$, thr.	0.944	0.951	0.974	0.944 (25)	0.0239
Transform mix	Gaussian	0.708	0.695	0.679	0.679 (50)	0.0079
Transform mix	Skew-t , $K = 1$	0.698	0.686	0.669	0.669 (50)	0.0080
Transform mix	Symmetric-t, $K = 1$, thr.	0.554	0.576	0.679	0.554 (30)	0.0095
Transform mix	Skew-t , $K = 5$	0.303	0.264	0.257	0.257 (50)	0.0210
Transform mix	Symmetric-t, $K = 5$, thr.	0.374	0.446	0.367	0.367 (50)	0.0131

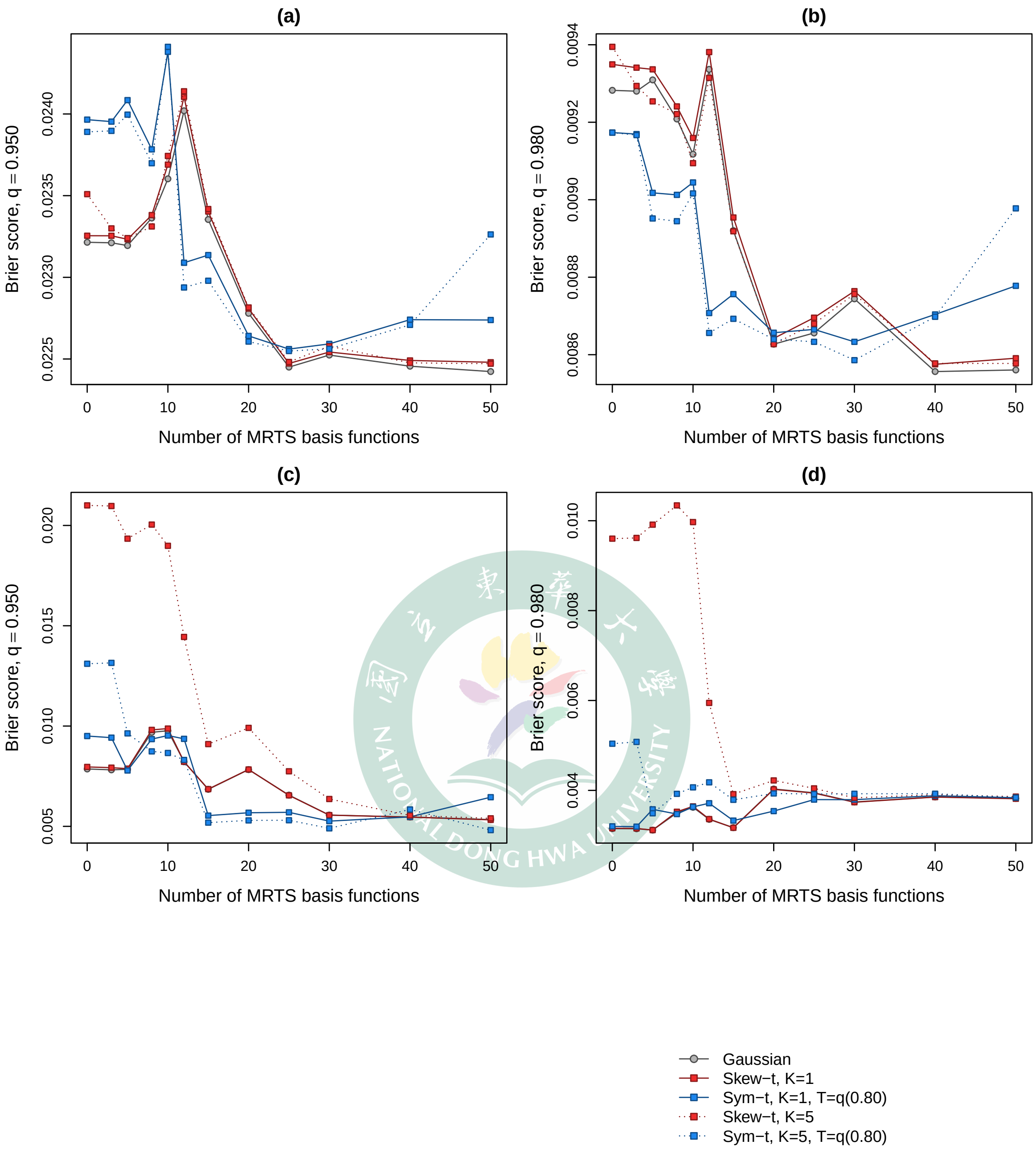


Figure 4: Brier score versus the number of MRTS basis functions K_{MRTS} on the two nonlinear-mean designs of (35) ($K_{\mathrm{MRTS}} = 0$ is the no-MRTS baseline; lower is better). Data-generating processes: (a,b) arc front; (c,d) transform mix. Both use 10 data sets on the full K_{MRTS} grid. Exceedance level $q = 0.95$ in (a,c) and $q = 0.98$ in (b,d).

3.5 Summary of the simulation study

Overall, the simulation study indicates that both extensions enlarge the usable model class but that neither uniformly dominates the STP baseline. The persistence experiment sharpens this conclusion into a statement about the prediction task itself. Even under highly persistent and long-memory dependence, with the temporal coefficients well identified, spatial hold-out prediction is insensitive to the autoregressive order; modest gains from richer temporal pooling, directionally consistent though never statistically significant, emerge only once the task requires extrapolating beyond the observed record. Whatever promise the AR(2) extension retains therefore lies in forecasting rather than interpolation. The MRTS experiments admit an equally structural summary. The mean term is inert when the non-stationarity lives in the second moment, and under genuine nonlinear mean structure it pays off only once the basis is fine enough to resolve the surface, converting a small fraction of the oracle headroom. Whether the same pattern holds on real environmental data is the question taken up in Section 4, where the AR(2) and MRTS variants enter as candidate options to be selected by held-out predictive scores rather than as default replacements for the AR(1) STP baseline.

4 Ozone data analysis

Morris, Reich, Emeric Thibaud, and Cooley (2017) introduced the spatio-temporal skew-t (STP) process to predict high-threshold ground-level ozone exceedances from U.S. EPA monitoring data. We return to the same July 2005 data, asking not whether a new model class is needed but whether enlarging the STP candidate class along the two axes of this thesis, the AR(2) temporal prior and the MRTS spatial-mean term, yields better held-out tail prediction. The comparison is therefore designed to identify the exceedance levels, if any, at which the enlarged candidate class improves on the STP baseline, rather than to select a single overall-best model.

4.1 Data

The response is the daily maximum 8-hour ozone concentration for the 31 days of July 2005 at U.S. EPA Air Quality System (AQS) monitoring sites, with the co-located Community Multi-scale Air Quality (CMAQ) model output as the sole covariate. Figure 5 summarizes the two data sources through the EPA design value, the fourth-highest daily maximum 8-hour average of the month. The AQS monitors are irregularly spaced, covering the eastern United

States and California densely but leaving large gaps in the interior west, whereas CMAQ provides complete gridded coverage of the conterminous United States. CMAQ reproduces the broad spatial pattern of the observations but not their level at individual monitors, which motivates using it as a regression covariate to be calibrated rather than as a substitute for the observations. Following Morris, Reich, Emeric Thibaud, and Cooley (2017), the regression component is $\mathbf{X}_t(\mathbf{s}) = [1, \mathrm{CMAQ}_t(\mathbf{s})]^\top$, so that each model acts as a statistical calibration of CMAQ against the monitor observations.

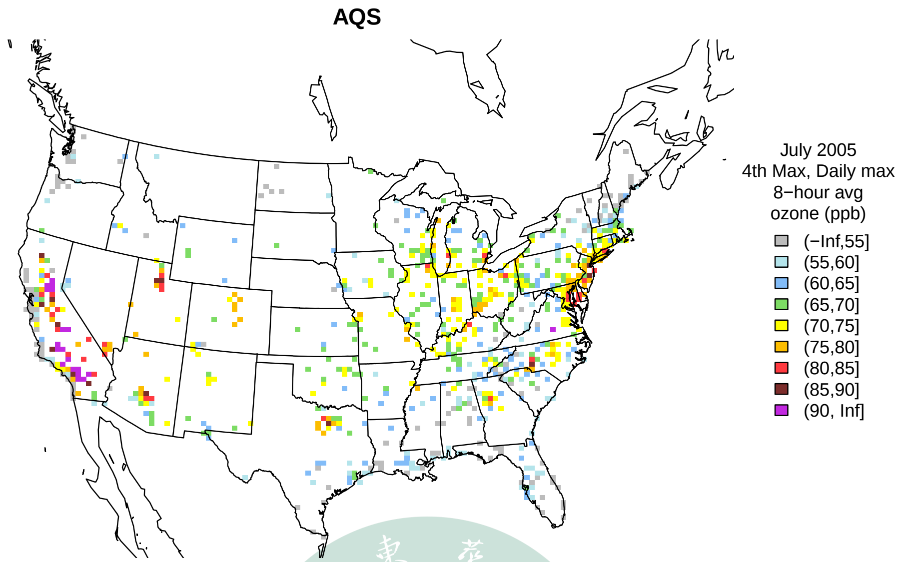
These data motivate the skew-t marginal directly. Regressing the observations on CMAQ at the $1{,}089$ sites above and pooling the residuals across sites and days, the QQ plot against the Normal distribution departs from the diagonal in both tails, with the upper tail by far the heavier (Figure 6); the largest standardized residuals reach eight standard deviations, far beyond what a Gaussian model can accommodate. Against the skew-t distribution with skewness $\lambda = 1$ and $a = 10$ degrees of freedom that Morris, Reich, Emeric Thibaud, and Cooley (2017) use for the same diagnostic, the plot straightens except for the few most extreme points. The residual ozone variation is thus heavy-tailed and right-skewed exactly in the region that exceedance prediction cares about, and a skew-t process is a natural candidate class for it.

4.2 Candidate models and validation design

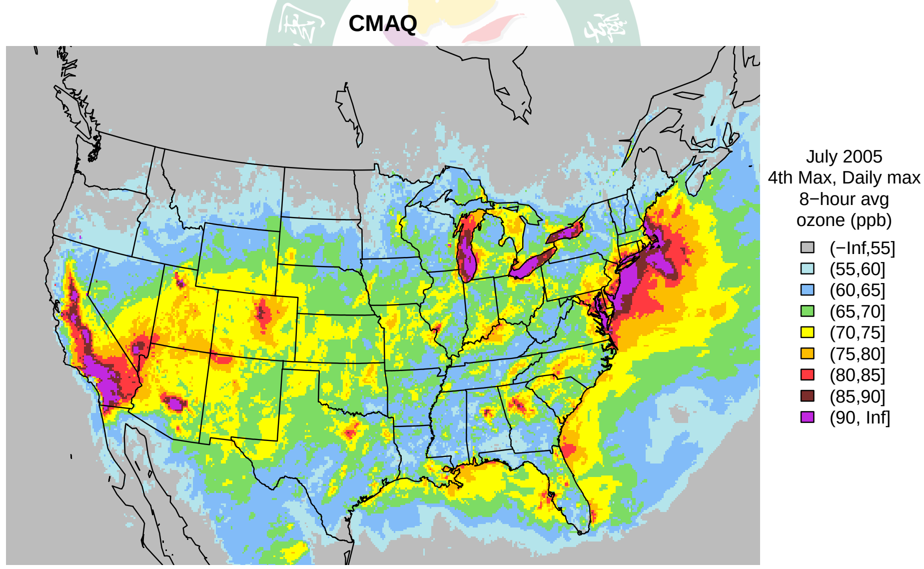
The baseline candidate set follows that of Morris, Reich, Emeric Thibaud, and Cooley (2017). We fit Gaussian and skew-t marginals over a range of partition sizes, $K \in \{1, 5, 6, 7, 8, 9, 10, 15\}$, and censor the response at three levels to contrast no, moderate, and high censoring, namely no censoring ($T = 0$), a moderate threshold ($T = 50$ ppb, the 0.42 sample quantile), and a high threshold ($T = 75$ ppb, the 0.92 sample quantile). For settings censored at $T = 75$ ppb, we use a symmetric-t marginal, as in the simulation study. Each setting is fit both with and without a latent time series.

On top of this baseline we add the two extensions developed in this thesis. The first replaces the AR(1) latent dynamics with the standardized AR(2) prior of Section 2.3; the second augments the regression component with MRTS basis functions, so that non-stationary spatial mean structure can be absorbed by multi-resolution thin-plate splines. Unlike the simulation study, which sweeps a grid of basis sizes, the ozone candidate set fixes $K_{\text{MRTS}} = 10$, so the MRTS comparison below is a pre-specified one rather than a best-over-K_{MRTS} selection.

Of the $1{,}089$ sites of Figure 5, the comparison draws on a stratified sample of 800, drawn to represent each region of the United States equally so that the interior west is not swamped by the dense eastern network. Models are evaluated by two-fold cross validation, with the 800



(a) AQS monitor observations



(b) CMAQ model output

Figure 5: July 2005 ozone design value (fourth-highest daily maximum 8-hour average, in ppb): (a) at the $1{,}089$ AQS monitoring sites missing at most 50% of the 31 days; (b) on the full CMAQ grid.

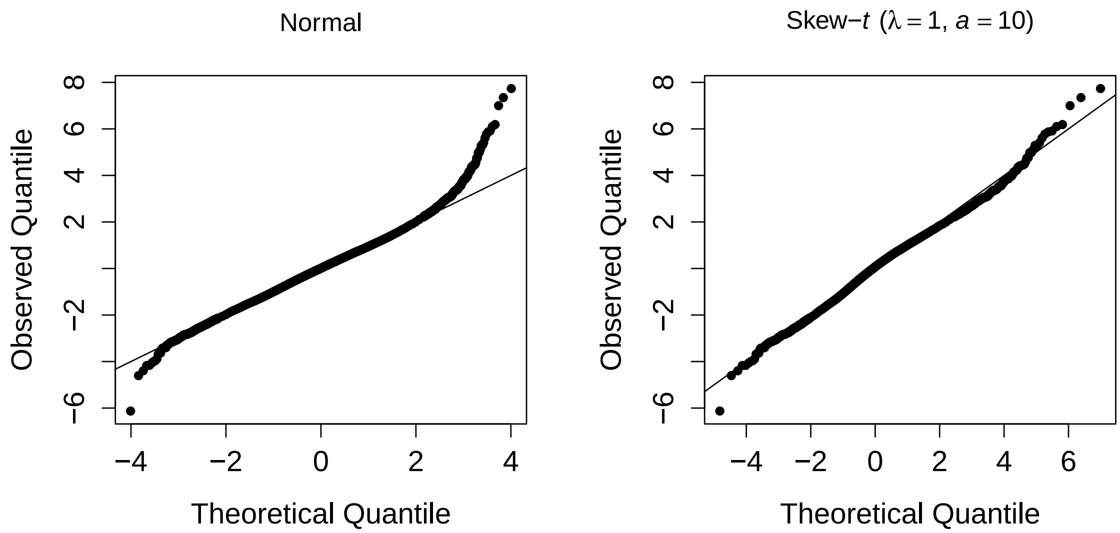


Figure 6: QQ plots of the pooled residuals from the linear regression of ozone on CMAQ at the $1{,}089$ AQS sites, against the Normal distribution (left) and the skew-t distribution with skewness $\lambda = 1$ and $a = 10$ degrees of freedom of Morris, Reich, Emeric Thibaud, and Cooley (2017) (right). Both reference distributions are standardized to zero mean and unit variance.

sites split at random into 400 training and 400 validation sites per fold, scored by the held-out Brier score of Section 3.2; robustness of the ranking to the split is examined with two further splits in Section 4.3. Each MCMC chain is run for $30{,}000$ iterations with a burn-in period of $25{,}000$ iterations.

4.3 Results

Across the upper tail, the proposed extensions appear among the best models at scattered exceedance levels rather than improving prediction uniformly, so no single extension, and no single setting, is best across the whole tail (Table 6). The scatter is nevertheless systematic. The MRTS extension leads at the 0.900 level, where the spatial mean still carries predictive signal; the AR(2) extension occupies both top ranks at 0.950 and returns at the deepest level, 0.995, where the best settings also shift toward high censoring and large partition sizes; the intermediate levels, 0.980 and 0.990, belong to the baseline family. The extensions thus appear as the leading model at several levels; how much statistical weight that ranking carries is assessed below.

This picture is reproducible on smaller training samples. Repeating the comparison under

Table 6: Top two performing models for predicting ozone exceedance of level L with Brier score.

Level L	Rank 1 model	Rank 2 model
0.900	**MRTS** skew-t , $K = 1$, $T = 0$, AR(1), $K_{\text{MRTS}} = 10$	skew-t , $K = 1$, $T = 0$, AR(1)
0.950	**AR2** skew-t , $K = 7$, $T = 50$	**AR2** skew-t , $K = 10$, $T = 50$
0.980	skew-t , $K = 6$, $T = 50$, AR(1)	skew-t , $K = 5$, $T = 50$, AR(1)
0.990	symmetric-t, $K = 9$, $T = 75$, AR(1)	skew-t , $K = 5$, $T = 50$
0.995	**AR2** symmetric-t, $K = 15$, $T = 75$	symmetric-t, $K = 9$, $T = 75$, AR(1)

Table 7: Top four Brier score models for the 300/100 split. Baseline denotes any other model carrying neither proposed extension; AR2 and MRTS denote the proposed temporal and spatial-mean extensions.

Level L	Rank 1	Rank 2	Rank 3	Rank 4
0.900	**MRTS** (0.041)	Baseline (0.041)	**MRTS** (0.041)	Baseline (0.041)
0.950	Baseline (0.020)	Baseline (0.020)	Baseline (0.020)	Baseline (0.020)
0.980	**MRTS** (0.008)	**AR2** (0.008)	Baseline (0.008)	**AR2** (0.008)
0.990	**AR2** (0.003)	Baseline (0.003)	Baseline (0.003)	Baseline (0.003)
0.995	Baseline (0.001)	Baseline (0.001)	Baseline (0.001)	**AR2** (0.001)

Table 8: Top four Brier score models for the 200/200 split. Baseline denotes any other model carrying neither proposed extension; AR2 and MRTS denote the proposed temporal and spatial-mean extensions.

Level L	Rank 1	Rank 2	Rank 3	Rank 4
0.900	Baseline (0.050)	**AR2** (0.050)	Baseline (0.050)	**MRTS** (0.050)
0.950	**AR2** (0.030)	**AR2** (0.030)	**AR2** (0.030)	Baseline (0.030)
0.980	**MRTS** (0.015)	**MRTS** (0.015)	**AR2** (0.015)	Baseline (0.015)
0.990	Baseline (0.008)	**AR2** (0.008)	**AR2** (0.008)	Baseline (0.008)
0.995	Baseline (0.004)	Baseline (0.004)	Baseline (0.004)	**AR2** (0.004)

two further train/validation splits, written $N_{\text{train}}/N_{\text{test}}$ for N_{train} training and N_{test} validation sites (the 300/100 and 200/200 splits; Tables 7 and 8), recovers the same qualitative ranking. The Baseline, AR(2), and MRTS families continue to share the top ranks across the tail, even as the exact best-ranked setting shifts with the split and the level; that shifting is expected, since only a handful of validation sites exceed the highest thresholds (Table 9 reports the exceedance counts) and the fine ranking there is sensitive to the held-out draw.

The ranking tables also overstate how much the tail separates these models, since the leading scores agree to the displayed precision. Table 9 therefore attaches an uncertainty statement to two matched comparisons, each isolating one extension while holding everything else fixed: the AR(2) model against its AR(1) counterpart, and the MRTS model against the same setting without basis functions. Following Section 3.2, the Brier difference is formed site by site and bootstrapped over validation sites, each site resampled together with all of its days to preserve within-site dependence ($B = 2000$, stratified by fold). For the AR(2) pair every interval covers zero under all three splits, so the added lag neither improves nor degrades held-out tail prediction. The MRTS pair differs. In the main comparison the difference is negative at every level and excludes zero at 0.95 and 0.99, and both intervals survive resampling contiguous geographic blocks in place of single sites, so the effect is not an artifact of treating nearby monitors as independent. Its size, however, is three to four tenths of a percent of the baseline score, which the smaller splits cannot resolve; the lone starred interval of the $200/200$ split, at the sparsest level with 27 exceedance events, fails that block check and is the kind of isolated borderline result chance alone produces. The two extensions thus part ways on the real data. The AR(2) prior leaves tail prediction unchanged, while the MRTS mean term buys a real but practically negligible improvement that only the largest validation set can detect, and the case for retaining both rests on the stability of the candidate class across splits and levels rather than on the size of any single gain.

Table 9: Paired held-out Brier differences for each extension against its matched baseline, $\Delta = \mathrm{BS}_{\mathrm{ext}} - \mathrm{BS}_{\mathrm{base}}$ in units of 10^{-3} (negative favors the extension), with 95% site-clustered bootstrap confidence intervals. The AR(2) pair is skew-t with K=7 and T=50; the MRTS pair is skew-t with K=1, T=0, a latent time series, and $K_{\mathrm{MRTS}} = 10$; the MRTS pair was not fit under the 300/100 split. n_{exc} counts the exceedance events behind each score; a star marks an interval that excludes zero.

		400/400		300/100		200/200	
Comparison	L quant.	$\Delta \times 10^3$ [95% CI]	n_{exc}	$\Delta \times 10^3$ [95% CI]	n_{exc}	$\Delta \times 10^3$ [95% CI]	n_{exc}
AR(2) − AR(1)	0.900	−0.15 [−0.38, +0.08]	2382	−0.03 [−0.53, +0.45]	241	+0.10 [−0.35, +0.54]	630
	0.950	−0.12 [−0.29, +0.04]	1198	−0.03 [−0.33, +0.32]	92	−0.06 [−0.42, +0.29]	319
	0.980	+0.03 [−0.08, +0.14]	477	−0.07 [−0.28, +0.13]	28	−0.09 [−0.25, +0.09]	129
	0.990	−0.06 [−0.14, +0.03]	239	+0.01 [−0.09, +0.10]	9	+0.02 [−0.09, +0.18]	56
	0.995	−0.03 [−0.08, +0.02]	120	+0.06 [−0.02, +0.16]	2	−0.03 [−0.08, +0.02]	27
MRTS − none	0.900	−0.03 [−0.13, +0.07]	2382	–	–	−0.07 [−0.48, +0.35]	630
	0.950	−0.08* [−0.14, −0.02]	1198	–	–	−0.08 [−0.34, +0.17]	319
	0.980	−0.04 [−0.07, +0.00]	477	–	–	−0.11 [−0.25, +0.00]	129
	0.990	−0.03* [−0.05, −0.00]	239	–	–	−0.05 [−0.12, +0.01]	56
	0.995	−0.01 [−0.02, +0.01]	120	–	–	−0.03* [−0.06, −0.00]	27

5 Discussion

This thesis asked not whether the spatio-temporal skew-t process predicts high-threshold exceedances well, which Morris, Reich, Emeric Thibaud, and Cooley (2017) already established, but where within it room for sharper upper-tail prediction remains. The temporal axis divides along the prediction task. Added memory leaves spatial interpolation unchanged, and not for want of signal: under persistent and long-memory dynamics the coefficients were recovered closely and the held-out scores still did not move, because with the full record observed at the training sites the daily cross-sectional information dominates. Only forecasting beyond the observed record rewarded richer temporal pooling, and then by margins never statistically significant. The spatial axis is favorable but small, the MRTS mean term is inert against second-moment non-stationarity, and the real improvement it bought on the ozone data was one only the largest validation set could detect. Two caveats bound these conclusions, the evidence covers a single month of a single pollutant, and the forecast evaluation used a single origin.

The implication extends beyond ozone, since the same target, a spatial probability statement about the upper tail at unmonitored sites, arises for any heavy-tailed, asymmetric environmental field regulated by a high threshold. Temporal order and mean flexibility are not free improvements to such a model, but neither are they idle. Rather than replacing the STP baseline, the standardized AR(2) prior and the MRTS mean enlarge it into a candidate class from which held-out tail scores can select the setting that fits the data at hand. Where each axis fell short marks where to go next: entering the MRTS basis with random coefficients, as in fixed-rank kriging, would place the flexibility where the simulation showed the structure to be, while a reparameterized z, longer records, and a rolling forecast origin would test whether the extrapolation gains are real.

A Appendices

A.1 Notation

Several letters carry more than one meaning in this thesis. Table 10 lists those reused symbols together to make the intended reading explicit; symbols not listed here are used with a single meaning.

Table 10: Symbols that reuse a letter or shape, with their distinct meanings.

Symbol	Meaning
$\phi_1,\ \phi_2$	AR(2) coefficients
$\phi^{(w)},\ \phi^{(z)},\ \phi^{(\sigma)}$	AR(1) coefficients of the knot-location, skewness, and scale processes
φ	Matérn spatial range
Φ	Standard normal distribution function
$\psi_i(\mathbf{s}),\ \boldsymbol{\psi}(\mathbf{s}),\ \boldsymbol{\Psi}$	Thin-plate spline kernel, its vector, and $n \times n$ matrix
γ_h	AR stationary autocovariance, constrained so $\gamma_0 = 1$
η	Matérn proportion of spatial variance, $\eta \in [0, 1]$
$\Gamma(\cdot)$	Gamma function (in the Matérn correlation)
λ	Skew-t skewness parameter
$\mu_1 \geq \cdots \geq \mu_n$	MRTS eigenvalues (of $\mathbf{Q}\,\boldsymbol{\Psi}\,\mathbf{Q}$)

Table 10 (continued)

Symbol	Meaning
$\mathbf{X}_t(\mathbf{s}),\ \mathbf{X}_t^*(\mathbf{s})$	Regression and MRTS-augmented covariate vectors
$\mathbf{R},\ \mathbf{r}$	MRTS low-order polynomial design matrix and a single-site row; $\mathbf{R}(\varphi, \nu, \eta)$ is instead the Matérn correlation matrix of (26)
K	Number of partition knots
K_{MRTS}	Number of MRTS basis functions
K_ν	Modified Bessel function of the second kind, order ν
$\sigma_t(\mathbf{s}),\ \sigma_{tk}^2$	Skew-t scale process
σ_ϵ^2	AR innovation variance
$\sigma_\beta^2,\ \sigma_\lambda^2$	Prior variances of $\boldsymbol{\beta}$ and λ
σ_g	Error scale of the nonlinear-mean designs
$Y_t,\ Y_t(\mathbf{s})$	Generic scalar latent AR process in the Yule–Walker and MCMC derivations; $Y_t(\mathbf{s})$ is instead the observed response at site $\mathbf{s}$
$z_{tk},\ z_{tk}^*$	Latent skewness variable and its PIT-transformed value; $\lvert z_t(\mathbf{s})\rvert$ in (1) is the half-normal skew term
h	Spatial distance $\lVert \mathbf{s}_1 - \mathbf{s}_2 \rVert$; also the temporal lag in the Yule–Walker context
$g(\mathbf{s})$	Fixed nonlinear mean surface
$g_j(\mathbf{s})$	j-th cosine field of the random effect
$v_t(\mathbf{s})$	Standard Gaussian (Matérn) process
$\mathbf{v}_k$	k-th eigenvector (column of $\mathbf{V}$)
n	Number of locations that build the MRTS basis
$n_s,\ n_t$	Number of sites; number of days

Table 10 (continued)

Symbol	Meaning

A.2 Stationary moments of the standardized AR(2)

We derive the Yule–Walker moments (13)–(14) used to standardize the AR(2) prior (11) to unit marginal variance. Let $\gamma_h = \mathrm{Cov}(Y_t, Y_{t-h})$ be the stationary autocovariances, constrained so that $\gamma_0 = \mathrm{Var}(Y_t) = 1$. Multiplying the recursion (11) by Y_{t-h} for $h \geq 1$ and taking expectations, and using that ϵ_t is independent of Y_{t-h} for the causal stationary solution, yields the Yule–Walker recursion

$$\gamma_h \;=\; \phi_1\,\gamma_{h-1} \;+\; \phi_2\,\gamma_{h-2}, \qquad h \geq 1. \tag{36}$$

Evaluating (36) at $h = 1$ and using the symmetry $\gamma_{-1} = \gamma_1$ with $\gamma_0 = 1$ gives $\gamma_1 = \phi_1 + \phi_2\,\gamma_1$, hence $\gamma_1 = \phi_1/(1 - \phi_2)$, and at $h = 2$ gives $\gamma_2 = \phi_1\,\gamma_1 + \phi_2$; together these are (13). Multiplying (11) instead by Y_t and taking expectations, with $\mathrm{E}[Y_t\,\epsilon_t] = \sigma_\epsilon^2$ because Y_t carries the current innovation, gives $\gamma_0 = \phi_1\,\gamma_1 + \phi_2\,\gamma_2 + \sigma_\epsilon^2$, and solving for σ_ϵ^2 under $\gamma_0 = 1$ recovers (14). The same normalization fixes the initial pair, which has unit variances and lag-1 covariance γ_1 and is therefore the stationary bivariate Gaussian (15).

A.3 AR(2) full conditional for the Gibbs-within-MH update

The AR(2) Markov structure factorizes the joint prior of $(Y_1, \dots, Y_{n_t})$ as

$$p(Y_1, \dots, Y_{n_t}) \;=\; p(Y_1, Y_2) \prod_{s=3}^{n_t} p(Y_s \mid Y_{s-1}, Y_{s-2}), \tag{37}$$

with $p(Y_1, Y_2)$ given by (15) and each conditional factor for $s \geq 3$ following (11). For an interior time point $3 \leq t \leq n_t - 2$, the value Y_t enters (37) through three factors, namely the self factor $p(Y_t \mid Y_{t-1}, Y_{t-2})$, the lag-1 factor $p(Y_{t+1} \mid Y_t, Y_{t-1})$, and the lag-2 factor $p(Y_{t+2} \mid Y_{t+1}, Y_t)$. The full conditional in the Gibbs-within-MH update of Y_t is therefore

$$\pi(Y_t \mid Y_{-t}, \mathbf{Y}) \;\propto\; L(\mathbf{Y} \mid Y_t) \cdot \underbrace{p(Y_t \mid Y_{t-1}, Y_{t-2})}_{\text{self}} \cdot \underbrace{p(Y_{t+1} \mid Y_t, Y_{t-1})}_{\text{lag-1}} \cdot \underbrace{p(Y_{t+2} \mid Y_{t+1}, Y_t)}_{\text{lag-2}}, \tag{38}$$

where $L(\mathbf{Y} \mid Y_t)$ is the spatial likelihood contributed by Y_t through the partition and Gaussian process structure of (1). The lag-2 factor is absent for $t = n_t - 1$ and both lag-1 and lag-2 factors are absent for $t = n_t$. At the head of the series the initial-pair density (15) plays the role of the self factor: Y_1 enters only the initial-pair factor $p(Y_1, Y_2)$ and the conditional $p(Y_3 \mid Y_2, Y_1)$, while Y_2 enters $p(Y_1, Y_2)$, $p(Y_3 \mid Y_2, Y_1)$, and $p(Y_4 \mid Y_3, Y_2)$; in each case the factors referencing a non-existent time index are simply dropped. Under a symmetric Gaussian random-walk proposal $Y_t^{(c)} \sim N(Y_t, s^2)$, with s^2 tuned to an acceptance rate near 0.3–0.4, the Metropolis–Hastings acceptance ratio is

$$R \;=\; \min\left(1,\; \frac{\pi(Y_t^{(c)} \mid Y_{-t}, \mathbf{Y})}{\pi(Y_t \mid Y_{-t}, \mathbf{Y})}\right), \tag{39}$$

in which all three prior factors must appear; omitting the lag-2 factor distorts the stationary distribution of the chain away from the AR(2) joint and produces a lag-2 autocorrelation that does not match the Yule–Walker value in (13).

A.4 Marginal covariance of the non-stationary-dependence design

The second MRTS design of Section 3.4 places the non-stationarity in the dependence through a mean-zero random effect built from two fixed cosine fields on the $[0, 10]^2$ domain. With centers $\mathbf{c}_1 = (0, 10)^\top$, $\mathbf{c}_2 = (7.5, 2.5)^\top$ and wavenumbers $\kappa_1 = \pi/10$, $\kappa_2 = \pi/5$,

$$g_j(\mathbf{s}) = \cos\big(\kappa_j \,\|\mathbf{s} - \mathbf{c}_j\|\big), \qquad j = 1, 2, \tag{40}$$

and the random effect superposes them with independent, daily-redrawn Gaussian weights,

$$u_t(\mathbf{s}) = \sum_{j=1}^{2} \xi_{tj}\, g_j(\mathbf{s}), \qquad \xi_{tj} \overset{iid}{\sim} N(0, \tau_j^2), \quad (\tau_1, \tau_2) = (5, 3), \tag{41}$$

independently across j and across days t. The response adds u_t to the one-knot skew-t model,

$$Y_t(\mathbf{s}) = 10 + \lambda\, \sigma_t(\mathbf{s})\, |z_t(\mathbf{s})| + \sigma_t(\mathbf{s})\, v_t(\mathbf{s}) + u_t(\mathbf{s}). \tag{42}$$

Marginalizing over the weights, $u_t(\mathbf{s}) = \sum_j \xi_{tj}\, g_j(\mathbf{s})$ has zero mean, $\mathrm{E}[u_t(\mathbf{s})] = \sum_j g_j(\mathbf{s})\, \mathrm{E}[\xi_{tj}] = 0$, and, by the independence of ξ_{t1} and ξ_{t2}, the low-rank covariance

$$\mathrm{Cov}\big(u_t(\mathbf{s}), u_t(\mathbf{s}')\big) \;=\; \sum_{j=1}^{2} \tau_j^2\, g_j(\mathbf{s})\, g_j(\mathbf{s}'). \tag{43}$$

The pointwise variance $\mathrm{Var}\big(u_t(\mathbf{s})\big) = \sum_j \tau_j^2\, g_j(\mathbf{s})^2$ depends on location, so the dependence is non-stationary while the marginal mean stays constant at 10. Because the weights are redrawn independently across days, u_t carries no temporal dependence, keeping this design orthogonal to the AR(2) experiment.

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